



Dividing Monomials and Polynomials

A ratio is a comparison of two quantities, often written as a fraction. If the quantities are represented by polynomials, the ratio is a rational expression.

Definition

A **rational expression** is a ratio whose numerator and denominator are polynomials, and whose denominator is nonzero.

► **Think About It** A rational expression has the form $\frac{p}{q}$, where p and q are polynomials and $q \neq 0$. The restriction $q \neq 0$ ensures that $\frac{p}{q}$ is defined.

Finding Restrictions on the Domain of a Rational Expression

To find the domain of a rational expression, find the values of the variables that make the denominator equal to 0. These values are excluded from the domain.

EXAMPLE 1

Find the domain restrictions for the expression.

A $\frac{x^4 y}{3xy^5}$

SOLUTION

$$3xy^5 = 0 \text{ when } x = 0 \text{ or } y = 0.$$

The domain restrictions are $x \neq 0$ and $y \neq 0$.

B $\frac{3x^3 - 7x}{4 - 3x}$

SOLUTION

$$\begin{aligned}4 - 3x &= 0 \\4 &= 3x \\\frac{4}{3} &= x\end{aligned}$$

The domain restriction is $x \neq \frac{4}{3}$.

C $\frac{x^2 + 10x - 5}{x^2 + 2x - 48}$

SOLUTION

$$\begin{aligned}x^2 + 2x - 48 &= 0 \\(x + 8)(x - 6) &= 0 \\x + 8 = 0 \quad \text{or} \quad x - 6 &= 0 \\x = -8 \quad \quad \quad x &= 6\end{aligned}$$

The domain restrictions are $x \neq -8$ and $x \neq 6$. ■

Simplifying Rational Expressions

Simplifying a rational expression is similar to simplifying a fraction.

Identify and divide out any factors you find in the numerator and the denominator. Be careful, though; both the numerator and the denominator must be written as products of factors. You can't divide out a factor in the denominator with a term that is being added in the numerator.

EXAMPLE 2

Find the domain restrictions and simplify the expression.

A $\frac{8x^2y^5}{24xy^7}$

SOLUTION

The domain restrictions are $x \neq 0$ and $y \neq 0$.

$$\frac{8x^2y^5}{24xy^7} = \frac{8xy^5 \cdot x}{8xy^5 \cdot 3y^2} \quad \text{Factor the greatest common factor from the numerator and the denominator.}$$

$$= \frac{\overset{1}{\cancel{8xy^5}} \cdot x}{\overset{1}{\cancel{8xy^5}} \cdot 3y^2} \quad \text{Divide out the common factor.}$$

$$= \frac{x}{3y^2} \quad \text{Simplify.}$$

B $\frac{8a^2 - 20a + 16}{4a}$

SOLUTION

To find the restrictions on the domain, set $4a = 0$ and solve.

The domain restriction is $a \neq 0$.

$$\frac{8a^2 - 20a + 16}{4a} = \frac{8a^2}{4a} - \frac{20a}{4a} + \frac{16}{4a} \quad \text{Divide each term of the polynomial by the monomial.}$$

$$= \frac{\overset{2a}{\cancel{8a^2}}}{\underset{1}{\cancel{4a}}} - \frac{\overset{5}{\cancel{20a}}}{\underset{1}{\cancel{4a}}} + \frac{\overset{4}{\cancel{16}}}{\underset{a}{\cancel{4a}}} \quad \text{Divide out common factors.}$$

$$= 2a - 5 + \frac{4}{a} \quad \text{Simplify.}$$

C $\frac{(x+1)(x+3)}{(x+1)(x+2)}$

SOLUTION

To find the restrictions on the domain, set $(x+1)(x+2) = 0$ and solve. The domain restrictions are $x \neq -1$ and $x \neq -2$.

$$\frac{(x+1)(x+3)}{(x+1)(x+2)} = \frac{\overset{1}{\cancel{(x+1)}}(x+3)}{\overset{1}{\cancel{(x+1)}}(x+2)} \quad \text{Divide out the common factor.}$$

$$= \frac{x+3}{x+2} \quad \text{Simplify.}$$

D $\frac{3x^2 + 16x - 12}{x^2 + 2x - 24}$

SOLUTION

To find the domain restrictions, set $x^2 + 2x - 24 = 0$ and solve.

$$\begin{aligned}x^2 + 2x - 24 &= 0 \\(x - 4)(x + 6) &= 0 \\x - 4 &= 0 \quad \text{or} \quad x + 6 = 0 \\x &= 4 \qquad \qquad x = -6\end{aligned}$$

The domain restrictions are $x \neq 4$ and $x \neq -6$.

$$\begin{aligned}\frac{3x^2 + 16x - 12}{x^2 + 2x - 24} &= \frac{(3x - 2)(x + 6)}{(x - 4)(x + 6)} && \text{Factor the trinomials.} \\&= \frac{(3x - 2)\cancel{(x + 6)}}{(x - 4)\cancel{(x + 6)}} && \text{Divide out the common factor.} \\&= \frac{3x - 2}{x - 4} && \text{Simplify.}\end{aligned}$$

E $\frac{5x^2 - 20x}{4 - x}$

SOLUTION

To find the restrictions on the domain, set $4 - x = 0$ and solve.

$$\begin{aligned}4 - x &= 0 \\4 &= x\end{aligned}$$

The domain restriction is $x \neq 4$.

$$\begin{aligned}\frac{5x^2 - 20x}{4 - x} &= \frac{5x(x - 4)}{4 - x} && \text{Factor the numerator.} \\&= \frac{5x(x - 4)}{-1(x - 4)} && x - 4 \text{ and } 4 - x \text{ are opposites. Rewrite } 4 - x \\&&& \text{in the denominator as } -1(x - 4). \\&= \frac{5x\cancel{(x - 4)}}{-1\cancel{(x - 4)}} && \text{Divide out the common factor.} \\&= -5x && \text{Simplify. } \blacksquare\end{aligned}$$

► **Think About It** To demonstrate that $x - 4$ and $4 - x$ are opposites, choose a value for x . Let $x = 7$.

Then $x - 4 = 7 - 4 = 3$ and $4 - x = 4 - 7 = -3$.

In general, binomial factors of the forms $(a - b)$ and $(b - a)$ are opposites of each other, and $\frac{a - b}{b - a} = -1$.

Interpreting Real-World Rational Expressions

Algebraic expressions are often used to represent real-world situations. It is important to interpret the meaning of each term of the expression in the context of the problem.

EXAMPLE 3

Canoes 1 and 2 can travel at a rate of 9 mph when there is no current.

Canoe 2 traveled 5 mi more than the distance (d) of Canoe 1.

The expression $\frac{d + 5}{9 + c}$ represents the time, in hours, it took for Canoe 2 to travel a particular distance when there is a current (c). Interpret the expressions $d + 5$ and $9 + c$ in the context of the problem.

SOLUTION

The distance formula is $d = rt$. When you divide this formula by r , you get the equation $t = \frac{d}{r}$.

The variable d in the expression $\frac{d + 5}{9 + c}$ represents the distance Canoe 1 traveled. Canoe 2 traveled 5 mi more than the distance of Canoe 1, so the distance traveled by Canoe 2 is $d + 5$.

Both canoes traveled at a rate of 9 mph with no current. The variable c represents the rate of the current in miles per hour. Canoe 2 traveled 9 mph plus the rate of the current, so the expression $9 + c$ represents Canoe 2's rate. ■